ROLE OF GENETIC MUTATION IN HUMAN DISEASE

DNA or deoxyribonucleic acid is a kind of molecular structure which transmits hereditary information from generation to generation. On the other hand, genes are particular portions of DNA found on a chromosome and encode proteins and RNA. Genes have a very significant impact on the physiology and organization of an organism. But at times, there is a change in the DNA structure. This change in the DNA structure is called genetic mutation.

GENETIC MUTATION

Mutation in genetics is responsible for causing different kinds of diseases. They can occur naturally due to mistakes during DNA replication, and also certain environmental conditions such as radiations and chemicals can cause mutation. Mutation will change protein synthesis and functioning of the cells, therefore changing normal functioning of the genes.

The objective is to learn about the contribution of genetic mutations to the development of diseases in humans and how changes in DNA sequences influence gene functions and cell activities. There are different types of genetic mutation based on the changes in the base pairs.

TYPES OF GENETIC MUTATIONS

1.POINT MUTATION

The point mutation occurs due to changes in the nucleotide base pairs in the DNA or RNA sequence. The point mutation is then subdivided into silent, missense or nonsense mutations. Silent mutation is characterized by the lack of changes in the protein because the new codon formed from the mutation codes for the same amino acid. The missense mutation leads to the development of a new protein with different amino acids. Nonsense mutation creates the stop codon thus the protein produced is defective.

2.INSERTION AND DELETION

Insertion and Deletion is a process that involves the insertion or deletion of one or more nucleotide bases in DNA sequence. Insertion and Deletion causes frameshift mutation depending on whether the number of the inserted or deleted nucleotide is multiple of three or not. Frameshift mutation creates an abnormal protein which is either shorter or longer than the normal protein.

3. CHROMOSOMAL ABERRATION

Chromosomal aberrations involve changes in structure or number of chromosomes.

Main chromosomal aberration includes deletion, duplication, inversion, translocation, aneuploidy. These aberrations can result in a genetic disorder like down syndrome, turner syndrome, klinefelter syndrome.

4. SPONTANEOUS MUTATION

A spontaneous mutation takes place naturally, without any external influences. This can happen because of mistakes in DNA replication, spontaneous lesions, transposable genetic elements. Mistakes in DNA replication take place during cell divisions and results in mutations of the genetic material. Spontaneous lesions damage DNA naturally. Transposable genetic elements are those DNA segments which can alter its position in genome or duplicate itself.

5. ENVIRONMENTAL FACTORS

Environmental factors influence the genetic material by damaging it physically or affecting its expression. Physical agents include ionizing radiation like X-rays, gamma rays, alpha rays, beta rays which break the bonds between DNA or RNA resulting in damage to the genetic code. Some chemicals also interfere with DNA replication and cause mutation.

GENETIC DISORDER

1.SICKLE CELL ANAEMIA

Sickle cell anaemia is a recessive genetic disorder that can be inherited by the offspring from the parent generation. Sickle cell anaemia can develop when a person gets sickle cell allele from both of his/her parents.

The disease is associated with the HBB gene. HBB gene codes for the beta globin chain of haemoglobin. There are two types of alleles of the haemoglobin gene. They are normal HbA allele and sickle cell HbS allele. If a person has HbS/HbS genotype then he/she will have sickle cell anaemia and if he/she has HbA/HbS genotype then the person has sickle cell trait.

The major cause of the disease is a point mutation in the beta globin gene. In this mutation, Glutamic acid (Glu) is substituted with valine (Val) at the sixth amino acid position in the beta globin chain. In this mutation, there is substitution of one nucleotide in the beta globin gene where the DNA codon of coding strand changes from GAG to GTG. Correspondingly, in the mRNA the DNA codon changes from GAG to GUG. Therefore, abnormal haemoglobin is produced, which is called HbS. When oxygen concentration is reduced in the body, the HbS forms sickle-shaped red blood cells. Sickle cell anaemia develops in babies at the age of 6 to 9 months. With age, the disease becomes severe since body starts producing more sickle shaped cells. The symptoms of the disease are anaemia and fatigue, pale skin, jaundice, swelling of hands and feet, frequent infection, delayed growth and puberty, shortness of breath, vision problem, fever. The management of the disease includes preventing pain, treating the symptoms, and complications of the disease. Treatment includes medicines and

blood transfusions. In the case of children, stem cell transplantation can treat this disease. The new gene therapy is being developed for sickle cell.

2.CYSTIC FIBROSIS

Cystic fibrosis is the recessive disorder caused by mutations of the gene which codes the protein called Cystic Fibrosis Transmembrane Conductance Regulator (CFTR). Those people who suffer from cystic fibrosis have two copies of this mutated gene. Carriers of the disease have only one gene with mutations, but generally, they have no symptoms of this disorder. The CFTR gene plays an important role in the regulation of chloride ions and water passage through cell membranes. Mutations of this gene lead to dysfunction of the transportation of salts and water resulting in the accumulation of thick mucus in different parts of the body.

Main organs that are affected by the disease are the lungs and the respiratory system because thick mucus is accumulated in the airways. This mucus provides good conditions for the growth of bacteria and the risk of infection of the lungs becomes higher. The most common bacteria which cause infections are *Pseudomonas aeruginosa* and *Staphylococcus aureus*. However, cystic fibrosis can affect such organs as pancreas, liver, and intestines. Thick mucus blocks ducts of the pancreas and prevents enzymes from passing to the intestine, resulting in poor digestion of food and nutrients.

Symptoms of cystic fibrosis include difficulties with breathing, coughing with mucus production, recurring infections of the lungs and sinuses, poor growth and problems with weight gain, fatty stool caused by poor digestion of fats. People who suffer from cystic fibrosis may have salty taste of their skin because of high concentration of salt in their sweat. The severity of symptoms depends on certain genetic mutations and affected organs.

Unfortunately, at the moment, there is no treatment of cystic fibrosis which can completely cure patients from the disorder. However, the symptoms can be controlled with proper treatment.

3.HUNTINGTON’S DISEASE

Huntington’s disease is a genetic disease passed from parents to children. The disease mostly affects the brain and nervous system. In this case of disease, the HTT gene is mutated because CAG repeats increase in HTT gene. The Huntington’s disease is an autosomal dominant disorder. A person will inherit one copy of the mutated gene from one of the parents. As a result, a mutated HTT gene causes the production of the huntingtin protein. Because of this mutation, the protein becomes abnormal and impacts the functioning of neurons causing damage to the brain cells. It symptoms include uncontrolled movements, difficulties in walking, problems with memory and behaviour, and problems with speech. Symptoms of the disease appear in adulthood and become worse as time passes by. There are

several methods of diagnosis including genetic testing that can detect an increase in the number of CAG repeats in the HTT gene. There is no cure for this disease, medications and support can help to supress the symptoms of the disease.

1. EFFECTS OF GENETIC MUTATION

The effects of mutations on the processes in the body may depends on the type of mutation, gene that is under mutations, and its importance for the organism. Sometimes mutations may cause no effects or can lead to development of genetic disorders.

Mutations may affect the structure of proteins and protein synthesis and changes the functioning of cells. For example, sickle cell anaemia is caused by the alteration of the structure of HBB gene that leads to changes in the structure of haemoglobin –which is responsible for oxygen transport in red blood cells. The result is the development of the abnormal shape of red blood cells that may block the blood vessels causing symptoms like pain, anaemia and deficiency of oxygen in the body.

Cystic fibrosis occurs due to mutations that change functioning of the gene that controls chloride ions and water movement through the cell membrane. As the result, the thick and sticky mucus forms especially in the lungs and digestive system causing infections and breathing problems.

Huntington's disease is caused by increased number of repetitions of CAG in HTT gene. Due to this protein becomes abnormal and causes damage of brain. Symptoms are uncontrolled movement problem with memory and speech.

2. DIAGNOSIS OF GENETIC DISORDER

There are different ways that can be used to diagnose genetic disorders to know whether there is the disorder and how the genetic changes occur. It helps in giving the patient the appropriate medical treatment.

The ways that can be used to check for the mutations in the particular genes is genetic testing. It involves analysing DNA to look for genetic changes related to the disorder. For example, genetic testing can identify mutation of the HBB gene causing sickle cell anaemia and also mutation of the CFTR gene causing cystic fibrosis.

Genetic testing is used to detect an increase in the number of CAG repeats in the HTT gene. Neurological examination helps to identify problems with movement, coordination, memory, and behaviour.

Blood tests can also be done to diagnose genetic disorders. For example, in sickle cell anaemia, blood tests can indicate the presence of anaemia and abnormal haemoglobin using either haemoglobin electrophoresis or HPLC. In cystic fibrosis, the sweat chloride test can be

performed to measure chloride levels in sweat. Increased levels of chloride would mean that there is cystic fibrosis and other genetic tests can follow.

Other ways of finding out whether genetic disorders are inherited include family history and genetic counselling.

3.MANAGEMENT OF GENETIC DISORDER

Management of genetic disorders depends on the particular disease, the degree of seriousness of the condition, and the symptoms experienced by the patient. Even though most of these diseases cannot currently be cured, there is a need to have proper treatment and management of the disease in order to control the symptoms and enhance the quality of life of the individual. Management of sickle cell anaemia include at relieving pain, hydration of the body, vaccines and medications to avoid any complications. Hydroxyurea will improve the level of foetal haemoglobin and lessen pain in many patients. In other cases, blood transfusion may be necessary. Stem cell transplantation will offer potential treatment for some patients while there are also gene therapy for some patient.

For cystic fibrosis, management involve improve lung functioning, prevent any infections, and improving digestion of patients. The patients will need airway clearance procedure to remove the mucus from the lungs, use antibiotics in order to kill bacteria, and also use enzymes to increase digestion of the food consumed. There are CFTR modulators which will help in improving functioning of the protein affected in some patients with the particular mutation.

Huntington's disease has no cure at present, but it can be managed with proper treatment and support. Medicines can help to control the uncontrolled movements and problems with memory and behaviour. Physiotherapy and regular care can help patients manage difficulties in movement and daily activities. Genetic counselling are also important for patients. Early diagnosis and proper care can help improve the quality of life of the patient.

Genetic counselling is also an important option in the management of the inherited diseases. The aim of genetic counselling is to help the patients to understand the ways of inheritance of the disease and genetic testing.

4.CONCLUSION

DNA sequence mutations are the changes that can influence the work of genes and proteins. Though some mutations do not bring any harm, some mutations cause hereditary diseases. Some of them are sickle cell anaemia and cystic fibrosis.

In the case of sickle cell anaemia, there is the mutation in HBB gene, causing change in haemoglobin structure. Cystic fibrosis is the disease in which the mutation in CFTR gene influences the transport of ions and water through the membrane. Huntington's disease is caused by increase number of repetition of CAG in HTT which affects the brain.

As it could be seen from the mentioned above mutations, genetic changes lead to some complications for the patients, shows the importance of genes for the normal work of the organism

The study of genetic mutations helps understand the reasons of genetic diseases, their ways of inheritance and ways of diagnosing and treating them. Genetic screening, early diagnosing, genetic counselling and adequate medical treatment can help to cope with them easier.

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